



30 August 2026

Dear Editors,

We are submitting the enclosed manuscript, "Air sampling in team congregate spaces for early detection of respiratory virus threats at the 2026 FIFA World Cup™" for consideration as a Research article in *Eurosurveillance*.

Respiratory infections are the leading illness category at major sporting events, yet surveillance still depends on athletes, team staff, and location-specific support staff recognising and reporting symptoms. That approach misses asymptomatic, pre-symptomatic, and unreported infections, so new infections are difficult to identify before they spread. We asked whether viral nucleic acid in the air of the hotel spaces occupied by a national squad could give team medical staff behaviour-independent warning during competition.

Working in partnership with the Canadian men's national soccer team, we ran continuous air sampling with same-day point-of-care testing across five host cities before and during the team's 2026 FIFA World Cup™ campaign. We tested 174 air filters and detected SARS-CoV-2, influenza A, and influenza B in shared team spaces even during the low-circulation Northern-hemisphere summer. Targeted metagenomic sequencing of the final samples additionally recovered human coronavirus OC43 and rhinovirus, which the point-of-care panel cannot detect. Detections clustered late in the tournament, and on three occasions coincided with events the team's medical staff independently recognised, including an influenza A signal the morning a player was withdrawn from training with a fever, and SARS-CoV-2 signals that fell after visibly ill hotel staff were excluded from the team spaces.

We believe this work is well matched to *Eurosurveillance* and its readership. It extends indoor air surveillance from the fixed community sites where it has been evaluated to a mobile cohort moving between cities, which is the situation faced by any national squad or delegation at a mass-gathering event. The closest published work is the year-long air sampling study in a Belgian daycare centre published in this journal in 2025, and our study applies the same principle under field conditions during competition. There are multiple reasons why the European readership of your journal would find this topical: the 2028 UEFA European Championship will bring 24 national squads to nine venues across eight cities in the United Kingdom and Ireland, and up to 12 of the 32 teams at the 2027 FIFA Women's World Cup™ will qualify through UEFA. Much of the evidence base on respiratory infection in elite athletes also comes from European, and particularly Nordic, cohorts. We propose that air sampling offers a behaviour-independent early-warning signal allowing intensive precautions to be applied selectively in response to data rather than continuously.

We confirm that this manuscript is original, has not been published previously, and is not under consideration by any other journal. A preprint is posted on medRxiv (DOI:10.64898/2026.08.16.26360542) and an interactive version of the same preprint is available at <https://dholab.github.io/team-canada-world-cup-2026/>. All authors have read and approved the submission and agree to be accountable for



the work. As co-lead and corresponding authors, Shelby L. O'Connor and I accept full responsibility for the content of the manuscript.

In the interest of full disclosure, Shelby L. O'Connor and I are managing partners of Pathogenuity LLC, a consultancy that advises on environmental monitoring for pathogens. We also hold Honorary professorial fellow appointments at the University of Melbourne, Australia. This work was supported by Inkfish LLC and Heart of Racing. These relationships did not influence the design, conduct, analysis, or reporting of the study, and all competing interests are stated in the manuscript.

The University of Wisconsin-Madison Institutional Review Board has determined that air sampling in congregate spaces does not constitute human-subjects research, as no specimens or identifiers are collected from any individual. Data and code are openly available in our repository, and sequencing reads are deposited in NCBI SRA under BioProject PRJNA1513008. A data availability statement is included in the manuscript and in the submission system. We also note that generative AI models were used as tools for literature retrieval, drafting, analysis, and visualisation, with all output independently verified and edited by the authors, as described in the manuscript's statement on the use of artificial intelligence tools.

Potential reviewers with relevant expertise are suggested below.

- Emmanuel André (emmanuel.andre@uzleuven.be). Senior author on the indoor air surveillance studies in Belgian community settings, including the daycare metagenomics study published in this journal, and on the work relating air pathogen detection to ventilation and carbon dioxide concentration.
- Olli Ruuskanen (olli.ruuskanen@utu.fi). Senior author on studies of respiratory viral infection risk in athletes, including air-travel-associated transmission.
- Maarit Valtonen (maarit.valtonen@kihu.fi). Leads work on respiratory viral infections in elite athletes at major winter events, including molecular point-of-care testing in national teams.

On behalf of our co-authors, I thank you for considering our manuscript. I am happy to provide any further information the editorial team may need.

Sincerely,

David O'Connor, Ph.D.
UW Foundation Professor
University of Wisconsin-Madison